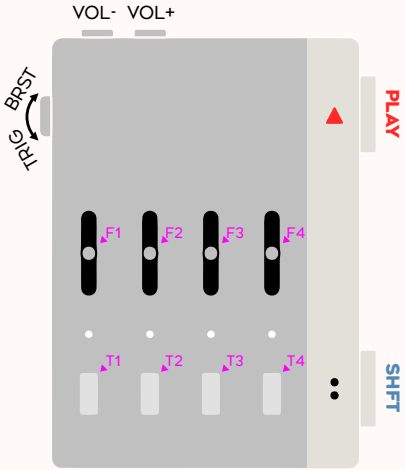
Wakes v0.9.0 **- Pg.1** - Welcome



• Hey! I'm **Adara**. Welcome to **Wakes**, a custom firmware for **Teenage Engineering**'s unreleased "**SP-1**" stem player. Wakes adapts code from **three** Eurorack modules into one device: **Plaits**, a macro-synthesizer, **Marbles**, a playful random sequencer, and **Yarns**, a signal converter. The modules were created by **Émilie Gillet**, otherwise known as **Mutable Instruments**.

• **CRITICAL:** Flashing the SP-1 with non-factory firmware is one-way. The SP-1 is adoptware; it is not manufacturer-supported, **and you cannot go back to the stock firmware if you flash it away**.

• Holding down **SHFT** for 1.5 seconds will turn the SP-1 **ON**, going to its first page, **Plaits**. Holding **SHFT** for 3 seconds while **ON** will turn the device **OFF**, or to **STANDBY** (showing battery level) if plugged in.

• While **ON**, any input while **SHFT** is held down will prevent shutdown, allowing you to work inside **SHFT** pages.

• Even if interrupted, holding down **SHFT** for 25 seconds will cause the face LEDs to flash quickly, notifying you that the SP-1 is about to perform an **emergency shutdown** in 5 seconds if you don't let **SHFT** go. Release and return to holding **SHFT** to get back to what you were doing. This is intended as an emergency backstop.

• At any point, even while **OFF**, tapping **SHFT** for a short moment will bring up a battery meter.

• While using **Wakes**, you'll be bouncing between **Plaits** to design sounds, and **Marbles** for sequencing, patterns and modulation.

• As you bounce back and forth, the face LED's will show you what's going on through a system of **glyphs**.

• Faders F1-F4 have a catch-up mechanism with the backend values, so moving them and swapping pages/engines doesn't create unexpected behaviours. For clarity, **follow the LED's** to know what the true parameters are.

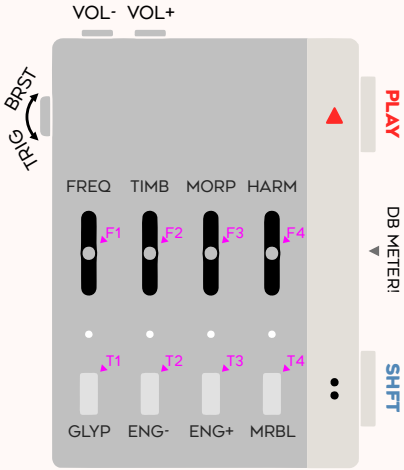
• Each module has a **SHFT** page containing advanced parameters and options that open up avenues for experimenting.

• Double-tapping **SHFT** enters a deep tinkering page called SETTINGS. There are two SETTINGS pages: One for Plaits, one for Marbles.

• Legend: ○ - Lit LED ● - Unlit LED

Wakes
A LOVE LETTER TO MUTABLE INSTRUMENTS
BY WORLDWAVE

Wakes v0.9.0 **- Pg.2** - Plaits - BASE



• **Wakes**' first page is **Plaits**: A synthesizer containing a multitude of sound engines representing different sound design techniques, with parameters attached to explorative macros. A **macro-synthesizer**.

• **Plaits** has four main faders: **FREQ** (F1), **TIMB** (F2), **MORP** (F3), and **HARM** (F4). Each of these controls several components of the selected synthesis engine.

• Plaits has a **low-pass gate**, controlled by an internally-routed **decay envelope**.

• **Read about what each engine is and what its parameters do in engines.cnfg. See more in pg. 9!**

• While in Plaits, the four side LED's will show a dB meter. Each LED represents 6dB of dynamic range.

• Generally speaking, **FREQ** will control the overall pitch of the sound. **TIMB** sweeps the spectral content from dark/sparse to bright/dense. **MORP** explores lateral timbral variations. **HARM** controls the frequency spread/overall balance of the spectrum.

• The rocker switch plays notes, triggering the lowpass gate. Press it down to send a single **TRIG**, pull it up to send a **BRST**!

• Engines are known by **glyphs**. Pressing **GLYP** (T1) will flash the glyph representing the selected engine. Hit **ENG+** (T3) or **ENG-** (T2) to browse engines.

• **PLAY** engages **Marbles' clock**, which is patched by default to trigger Plaits and give it musical notes.

• Press **MRBL** (T4) to go to **Marbles**.

Plaits - **SHFT**

• Hold **SHFT** to go to **Plaits SHFT:**

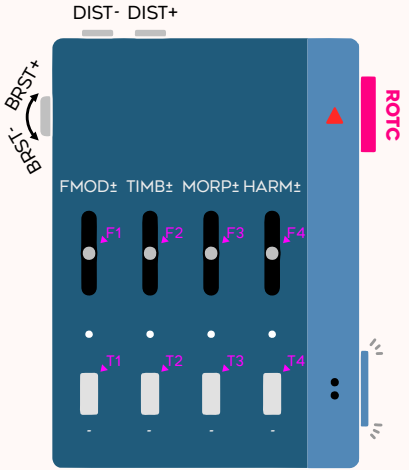
• Each fader here (F1-F4) represents a **modulation depth control** for incoming control signals. By default, Plaits' **decay envelope** is assigned to all of the four parameters. Use the faders to modulate Plaits' macros! Fun!

• **BRST-** and **BRST+** select **BRST**'s rhythmic subdivision: **1/1** through **1/128**.

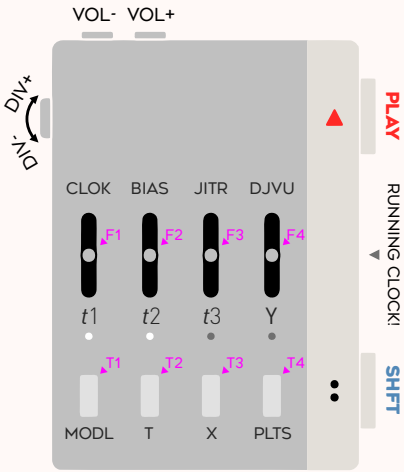
• **DIST-** and **DIST+** control the depth of a **master softclipper**. Loud and crunchy!

• Holding down any of T1-T4 for 2 seconds **Unpatches** Marbles modulators from that parameter.

• Holding down **PLAY** for 3 seconds **Rips Out Plaits' Cables**, returning it to its default configuration (First engine in your stack, all faders and attenuverters on neutral values, no incoming modulations).



Wakes v0.9.0 **- Pg.3** - Marbles - *t* Side (left)



• **Marbles** has two sides: ***t*** and **X**. **Glyphs** show which side you are on: ○○●● for ***t*** and ●●○○ for **X**. Each side has **3** outputs: ***t*1**-***t*3**, **X1**-**X3**. An extra output **Y** is offered.

• Moving F1-F4 on either page shows the fader values briefly, before returning to showing the individual ***t*** and **X** outputs.

• ***t*** is a random gate generator. It emulates a player following a stable metronome.

• **CLOK** (F1) controls BPM (default: 120).

• ***t*2** is **always** a stable clock: 50% on, 50% off. **BIAS** (F2) controls the distribution of gates coming out of ***t*1** and ***t*3**.

• **JITR** (F3) controls how chaotically gates land on ticks. Accurate, to super drunk.

• **DJVU** (F4) controls the probability of repeating patterns. At 50% it's a locked loop. Down from 50% adds random notes. Up from 50% reorders memorized notes.

• **MODL** (T1) selects **BIAS**' distribution model for ***t*1** and ***t*3** activity:

○●●○-cointoss: gates distribute evenly.

●○○●-clusters: random pairs of clock subdivisions appear at each side.

○●○●-drums: **BIAS** steps through drum pattern tables containing rhythms.

●○●○-indpcoin: each side tosses its own coin. ***t*1** and ***t*3** can coincide.

○○●○-divider: **BIAS** selects from fixed subdivision pairs. **8:1** >**1:1** > **1:8**.

○●○○-threestates: a three-way choice per clock tick: ***t*1**, ***t*3** or neither.

• **DIV-/DIV+** multiply the clock: **¼**, **1**, **4**.

• Pressing T2 shows you the current page.

• Press **X** (T3) to go to the **X** side of Marbles, or **PLTS** (T4) to go back to **Plaits**.

• Hold **SHFT** to go to ***t* SHFT:**

• The four buttons (T1-T4) select **Plaits modulation targets** for ***t*1**-***t*3**, or for **Y**. The mod targets are (in order): ●○○○-**TRIG**, ●●○○-**LEVL**, ○●●●-**FM**, ●○●●-**TIMB**, ●●○●-**MORP**, ●●●○-**HARM**, ●●●●-**none**. For **Y**, ●○○○ routes to **FREQ**.

• **GTLT** (F1) scales ***t*1** and ***t*3**'s max outputs.

• **LNTH** (F2) determines the length of on-states for ***t*1** and ***t*3** gate signals.

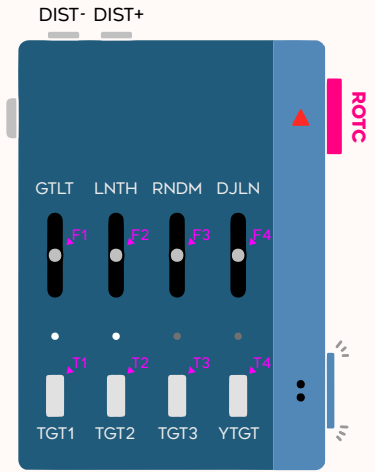
• **RNDM** (F3) determines gate length randomization for ***t*1** and ***t*3**.

• **DJLN** (F4) sets the maximum **DJVU** loop length: **1-8**, **10**, **12**, **14** and **16** beats.

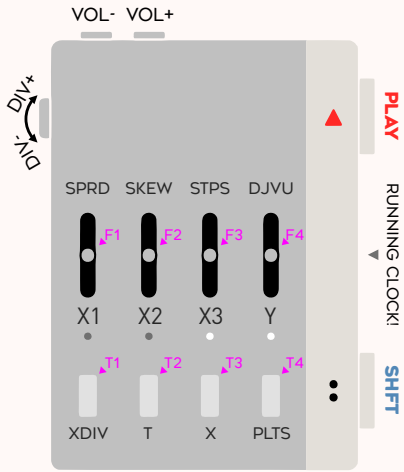
• Holding down any of T1-T4 for 2 seconds **Unpatches** any Plaits destinations.

• Hold **PLAY** for 3 seconds to **Rip Out Marbles' Cables**, returning it to its default state (***t*2** > **TRIG**).

Marbles - *t* **SHFT**

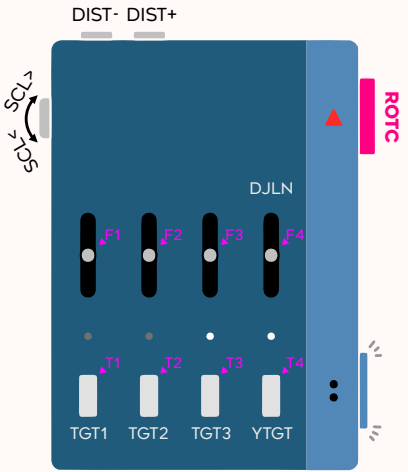


Wakes v0.9.0 **- Pg.4** - Marbles - X Side (right)



- **X** is a random modulation generator. Its three outputs, **X1**-**X3**, are clocked by the outputs of Marbles' ***t*** side, and generate random control signals that can vary from **Sample & Hold** style steps to **smooth, meandering modulations**. It can modulate Plaits' parameters, or be quantized to play musical intervals, creating melodic sequences, as well as enabling many other ideas.
- **Y** shows up on this page as well. It's a smooth, meandering full-range random source that's clocked at **1/8th** the rate of ***t2*** by default. **Y** never obeys **DJVU**, she's wild like that.
- **SPRD** (F1) spreads the concentration of signals around the central point.
- **SKEW** (F2) skews the average signal intensity up or down.
- **STPS** (F3) smooths or tightens the vertical steppiness of the control signals. Below-centre, the signals have random intensities. Above-centre, it **quantizes X to a musical scale**.
- **DJVU** (F4) is the same as on the ***t*** side. Both sections ***t*** and **X** are controlled by the same memory.
- Press **XDIV** (T1) to choose the intensity diversity among the outputs: **identical**, **bump** (**X2** is strongest) or **tilt** (**X1** is weakest, **X3** is strongest).
- Press ***t*** (T2) to go back to ***t***.
- Press **PLTS** (T4) to go back to **Plaits**.

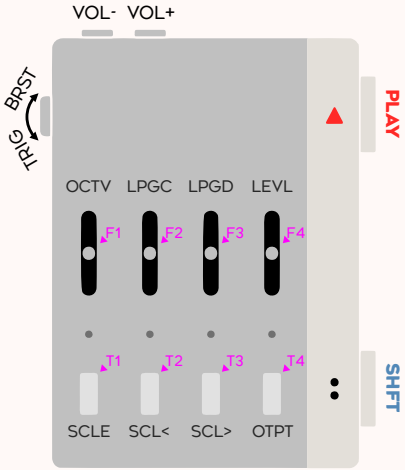
Marbles - X **SHFT**



Hold **SHFT** to go to **X SHFT:**

- The buttons (T1-T4) select **Plaits modulation targets** for **X1**-**X3**, or for **Y**. The targets are ○●●●-**FM**, ●○●●-**TIMB**, ●●○●-**MORP**, ●●●○-**HARM**, ●○○○-**FREQ**, ●●○○-**LEVL**, ●●●●-**none**. **FREQ** is the default input for musical tunings, whilst **FM** is full-range for experimentation.
- **DJLN** (F4), like in ***t***, controls the length of the **DJVU** sequence. **DJVU** and **DJLN** are the same across the two sides.
- **SCL<**/**SCL>** select from a range of scales to **quantize** into, when **STPS** is above its midpoint. **You can customize your available scales**, review **Pg.6** and **Pg.9**!
- Holding down any of **T1**-**T4** for 2 seconds **Unpatches** any Plaits destinations.
- Hold **PLAY** to **Rip Out Marbles' Cables**, returning it to its default state.

Wakes v0.9.0 **- Pg.5** - Plaits - **SETTINGS**



• Reach Plaits' **SETTINGS** page by double-tapping **SHFT** while in the Plaits module. Single-tapping **SHFT** will return you to where you were.

• **OCTV**'s (F1) default value is 100%. Just below max, **FREQ** sweeps full octaves of the root note. Below that, it locks Plaits' available frequencies to two octaves around the root note. Choose the range to play by moving the fader up/down. **FREQ**'s bottom value is full-range, but omits the C4 centre-detent, for the freaks out there.

• **LPGC** (F2) lets you control the **color** of the **low-pass gate**, which is composed of a **low-pass filter** and an **amplifier**. Lower values mix in more of the low-pass filter, making the **internal decay envelope** and **LEVL** sound duller in tone. Higher values use more of the amplifier, making the sound brighter.

• **LPGD** (F3) lets you change the length of the **decay envelope**. Higher values set longer lengths, making triggered notes take longer to decay. Any self-patching from the **SHFT** attenuverters will also be longer. Shorter values sound snappier and quieter overall.

• **LEVL** (F4) offsets **Plaits**' filter and amplifier. Moving it fades the sound in and out. The color of this fade is affected by **LPGC**. Setting **LEVL** to maximum will effectively **bypass the low-pass gate**, allowing **Plaits** to **drone**. Some Plaits engines will react beautifully to modulating **LEVL**.

• **Wakes** includes a **smart Plaits quantizer** feature. It lets you noodle around with notes on a scale by **moving FREQ**, without Marbles' modulation. It's set to ●●●●-off by default. Activate it by pressing **SCL>** (T3). Deactivate it by pressing **SCL<** (T2) until the glyphs show ●●●●-off again. Once activated, it first disables any Marbles routings going into **FREQ**. Then, for most scales, it samples the previously-selected **FREQ** parameter to use as the fundamental tuning/root note of the chosen scale. Then, it takes over **FREQ** and quantizes it to the selected scale.

• **SCLE** (T1) flashes the currently-active **quantizer scale**'s **glyph**. Scales are the same as Marbles', refer to the Marbles SETTINGS manual page for their glyphs.

• **Plaits** features an extra output, **AUX**, containing **additional content from each engine**. **Wakes** gives you **OTPT** (T4) on this page to select from different output configurations: ○●●●-**OUT**, ●○●●-**AUX**, ●●○●-**OUT+AUX**, and the **Wakes**-unique ●●●○-**OUT×AUX**.

Plaits Tips

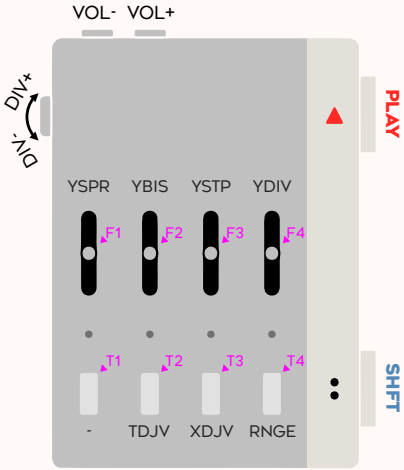
• Plaits is designed to be really breakable. Experiment with fast BRST, fast clocks, lots of mod targets. **Just go crazy with it**, and see what wilderness you end up in. When in doubt, **ROTC**.

• **FM** is a legacy Plaits input containing exp-FM for Plaits. It's used for atonal modulations.

• You can **TRIG** while **LEVL** is open ;)

• Combine the **quantizer** with **OCTV** to lock your scale's notes to a 2-octave range.

Wakes v0.9.0 **- Pg.6** - Marbles - **SETTINGS**



• Reach Marbles' **SETTINGS** page by double-tapping **SHFT** while in the Marbles module. Single-tapping **SHFT** will return you to where you were.

• **YSPR** (F1) lets you control **Y**'s spread, same way you do for X.

• **YBIS** (F2) lets you control **Y**'s randomization skew towards higher or lower signal strengths.

• **YSTP** (F3) lets you control its overall steppiness. Lower values are more sinusoidal. Central values are linear, triangular. Higher values are slewed squares, up to an even sample & hold.

• **YDIV** (F4) lets you choose at which subdivision of the clock Y is clocked. This lets you accelerate it to higher values if you want.

• **TDJV** (T2) and **XDJV** (T3) toggle **DJVU** for the ***t*** or **X** sides on or off. **DJVU** is on for both by default. If you toggle it off for one of the sides, that side won't be affected by **DJVU**'s looping algorithms.

• **RNGE** (T4) sets the control signal strength range for the **X** and **Y** outputs. The options are: ○●●● - 0-20%, ○○○●-0-50%, ○○○○-±50%, ○●○●-**INTL**. **INTL** is the default - a **Wakes**-unique upside of combining the two modules in a closed circuit: It decides your signal strength range selection for you, per-output, automatically, and with the target Plaits engine in mind. That said, if you still want to play around with unexpected results, the other modes remain available.

Scales:

• There are 7 default scales Wakes offers for quantizing the **X** outputs or **Plaits' FREQ** (F1) fader. When you press **SCL<** or **SCL>** on Marbles' **SHFT** page, their **glyphs** will flash on the face LEDs as you cycle through the list.

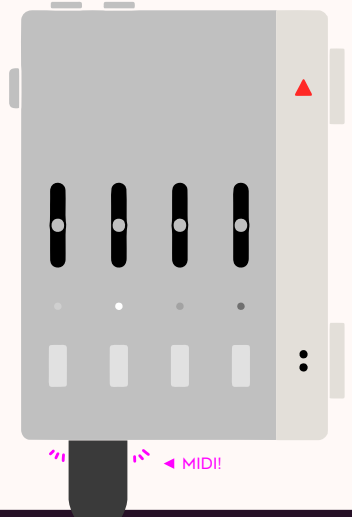
• Available Scales: ○●●●-major, ○○●●-minor, ○○○●-pentatonic, ○○○○-pelog, ●○○○-bhairav, ●●○○-shri, ●●●○-wholetone.

• If you choose one of these scales, then move **STPS** (**X** F3) above-center, then send one of the **X** outputs into **FREQ**, Marbles will tell Plaits to play notes on that scale. The overall tuning is still determined by Plaits' **FREQ** fader, though.

Marbles Tips

• You can combine multiple Marbles outputs into one Plaits destination. Using **Unpatch** disconnects only that specific output from its destination.

Wakes v0.9.0 - Pg.7 - MIDI & USB Audio



- Wakes supports **MIDI IN and Audio OUT** over the SP-1's **USB-C** port. Connect your SP-1 to a **USB host**, and it should pop up as a **MIDI** and **USB audio** device. The default MIDI channel is set to omni mode.
- Wakes automatically routes pitch and MIDI notes into **Plaits**, and routes MIDI clock into **Marbles**.
- By default, a MIDI CC behaves like a second hand on the physical faders, picking parameters up depending on which was last moved.
- You can edit advanced MIDI handling (such as **velocity** and **aftertouch**) and CC bindings (you can change engines via CC!) by editing **midi.cnfg**.
- Select USB audio handling and speaker options by editing **audio.cnfg** file in the device's onboard storage.

PLAITS - MIDI BEHAVIOURS

- Plaits will now receive **pitch**, **pitchbend (±2 default)**, **gate**, and **MIDI CC** information from incoming MIDI signals. This bypasses the quantizer.
- **FREQ**'s centre-detent will now expand to 10% making it easier to lock onto correct MIDI pitch. Moving **FREQ** will now transpose Plaits in relation to incoming MIDI notes.
- Plaits is a **monophonic** voice. It does not support polyphony except for **Chords**, **String** and **String Machine.** Those have distinct harmonic behaviours that are not traditional polyphony. **midi.cnfg** is set to legato mode with a portamento by default.
- **Plaits' MODEL** input is now exposed via MIDI CC, letting you select engines from the host. Caution - high CPU.

MARBLES - MIDI BEHAVIOURS

- **Marbles** will sync to external MIDI clocks until MIDI is disconnected. MIDI START is now the same as pressing **PLAY**.
- Marbles' **CLOK** (***t*** F1) fader now **selects a ratio** of the incoming MIDI clock which Marbles plays on. The ratios are: 1/4, 1/3, 1/2, 2/3, 1 (at centre), 3/2, 2, 3 or 4.
- Marbles' **DIV+** and **DIV-** rocker switch controls multiplying ranges of that subdivision: ○●●●-1/4, ●○○●-1, and ●●●○-4.
- Wakes compares incoming MIDI note timing with Marbles' clock, and runs Marbles that much ahead, so Marbles' steps land with your notes and your DAW needs no clock offset (leave it at 0).

DEFAULT MIDI CC LIST

Plaits

3-TIMB, 9-MORP, 14-HARM, 15-FREQ, 20-FMOD±, 21-TIMB±, 22-MORP±, 23-HARM±, 24-LPGC, 25-LPGD, 26-LEVL, 102-OCTV, 105-**MODEL**, 64 - Sustain pedal

Marbles

t

27-CLOK, 16-BIAS, 85-JITR, 17-LNTH, 86-RNDM 89-GTLT

X

28-SPRD, 29-SKEW, 30-STPS

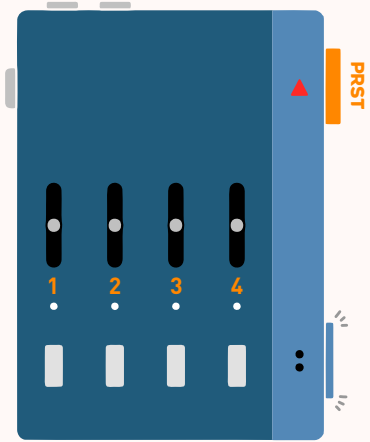
Y

18-YSPR, 19-YBIS, 87-YSTP, 104-YDIV

DJVU

31-DJVU, 103-DJLN

Wakes v0.9.0 - Pg.8 - Storage and PRST



STORAGE

• **Wakes** features onboard storage, which allows users to store and retrieve files for certain features.

• **First-time users:** The SP-1 stores samples in a 4GB eMMC card. Wakes needs to format it to operate. This will clear any data in that partition if you haven't backed it up (you probably don't need to care about what's there by default).

• When you first go to flash Wakes, you have a choice: **wakes-sp1.bin** or **wakes-sp1-fresh.bin**. If you've never flashed the **fresh** one before, use **fresh** - It'll format your eMMC card and prepare Wakes' folders, once per flash. Unless indicated otherwise, it's not necessary to flash **fresh** again, just use **wakes-sp1.bin** to preserve Wakes' storage. You'll know when you should flash **fresh** again.

• Wakes will behave like a removable mass-storage card reader while in its **STANDBY** state (**OFF** and USB is plugged in). While storage activity is ongoing, you'll see a dancing pattern on the face LED's. To protect files, you can't turn the SP-1 **ON** while storage activity is ongoing. If you want to unplug your SP-1 from your host, wait until the lights are fully dark before doing so, otherwise you risk file corruption. If the host is just a MIDI device, don't worry about unplugging abruptly; the drive doesn't get mounted in the first place.

• When Wakes is turned **ON**, it disconnects its storage from the host. To manipulate stored files, Wakes must be turned **OFF**. If you've ejected Wakes' storage while plugged in, you'll need to replug the SP-1 to your host for the drive to reappear.

• It's good to periodically save backups.

PRST - SAVESTATES

• **v0.8.0** also introduces **PRST**, allowing you to save and retrieve **full Wakes patches** that you've worked on as **.prst** files.

• Saved **PRST**s contain **every Wakes setting**, except for **VOL** and **DJVU**'s stored loop, in .ini files with a **.prst** file extension.

• When turned **ON**, Wakes always loads into the last-loaded **PRST** slot. It loads all slots from storage into RAM on boot. Effectively, while using Wakes, you are **always operating within one of the slots**.

• Wakes saves its state to the **currently selected PRST** slot **only** once turned **OFF** or to **STANDBY**. Moving to the next slot will **not** save Wakes' current state, it'll overwrite it.

• **30s emergency shutdowns skip saving.**

PRST - BROWSER

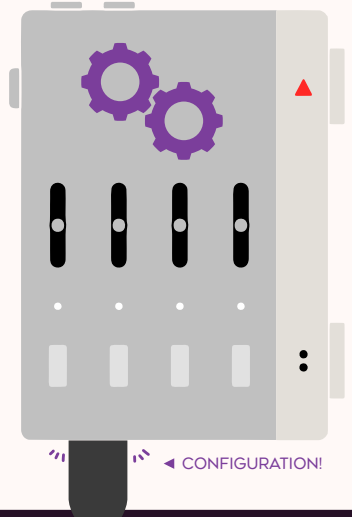
• While holding **SHFT**, tap **PLAY** once (instead of holding it to **ROTC**) to briefly enter the **PRST** browser, showing which slot you are currently on via a blinking LED.

• While the animation is still playing, Tap **PLAY** again quickly to flip to the next slot.

• Letting go of **SHFT** or waiting a short moment takes you out of the **PRST** browser.

• You can manipulate .prst files in Wakes' storage. Plug in your SP-1, go to <SP-1>:/WAKES/PRST and you will find folders representing the four **PRST** slots. The first file in alphabetical order will be the one Wakes loads. Any other files are ignored. Leaving the folder empty will load defaults. If you rename .prst files, keep to simple characters.

Wakes v0.9.0 **- Pg.9** - Advanced **CNFG**



CONFIGURATION FROM STORAGE

• Wakes **advanced configuration** can be done via **.cnfg** files found in the onboard storage when you plug it into your PC:

- WAKES
  - CNFG
    - audio.cnfg
    - engines.cnfg
    - logs.cnfg
    - midi.cnfg
    - scales.cnfg

• **.cnfg** files are defined in **.ini** format, and contain comments explaining what each setting does and what subformats they might use.

• Wakes reads its **.cnfg** files when it's turned **ON**. Missing files are replaced with defaults. Usually, bad **.cnfg**s are ignored.

PLAITS ENGINES CONFIGURATION

• **engines.cnfg** lets you configure your engine stack! Test, curate and rearrange your sound engines for a perfect selection.

• **engines.cnfg** also contains a reference for current and future **Plaits** engines offered by **Wakes**. Read the ENGINES section to learn about each engine, each parameter it offers, and what's on the AUX output that you can optionally use through Plaits' SETTINGS page.

• If there is a problem with your **engines.cnfg**, Wakes will skip the whole thing and generate the default engine stack (the original 16 Plaits engines).

SCALES

• Wakes lets you configure the scales used by Plaits' quantizer and Marbles' X **STPS** quantization, via **scales.cnfg**.

• scales.cnfg uses the **Scala .scl format** to define scales. It features instructions on how to use .scl to define your own scales, a link Scale Workshop where you can create & test scales, and a **reference scales wiki** for you to copy into your slots and use.

• You'll need musical theory experience to create scales that you can't find online.

• Wakes has 8 scale slots, including "no scale applied". This means that **scales.cnfg** features **7 slots** for you to play with. If you define more than 7 slots, extras get dropped. When in doubt, turn on logging, or ask Adara to help.

LOGGING

• WAKES/LOGS contains **.logs** files. Each session (from **ON** to **OFF/STANDBY**) will create a logfile with a session number as its name, for example **9001.logs**. See how many Wakes sessions you can get out of your SP-1!

• **logs.cnfg** allows you to configure granularity for the device's logging. If you encounter bugs, Adara might ask you to turn some of these on and send her one of your **.logs** files.

• Only the last 5 sessions are kept logged. Logs are written to the device's storage every 5 seconds. New logs rotate old logfiles automatically at every **ON**.